

A REPEATED POWER TRAINING ENHANCES FATIGUE RESISTANCE WHILE REDUCING INTRASET FLUCTUATIONS

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ABSTRACT

Oliver, GS, Julio, TF, Moras, G, José Luis, AS, and Alberto, MV. A repeated power training enhances fatigue resistance while reducing intraset fluctuations. *J Strength Cond Res* XX(X): 000–000, 2018—The present study analyzed the effects of adding an upper-body repeated power ability (RPA) training to habitual strength training sessions. Twenty young elite male basketball players were randomly allocated into a control group (CON, $n = 10$) or repeated power group (RPG, $n = 10$) and evaluated by 1 repetition maximum (1RM), incremental load, and RPA tests in the bench press exercise before and after a 7-week period and a 4-week cessation period. Repeated power group performed 1–3 blocks of 5 sets of 5 repetitions using the load that maximized power output with 30 seconds and 3 minute of passive recovery between sets and blocks, respectively. Between-group analysis showed substantial greater improvements in RPG compared with CON in: best set (AP_B), last set (AP_L), mean power over 5 sets (AP_M), percentage of decrement, fluctuation decrease during AP_L and RPA index (AP_{Lpost}/AP_{Lpre}) during the RPA test (effect size [ES] = 0.64–1.86), and 1RM (ES = 0.48) and average power at 80% of 1RM (ES = 1.11) in the incremental load test. The improvements of AP_B and AP_M were *almost perfectly* correlated. In conclusion, RPA training represents an effective method to mainly improve fatigue resistance together with the novel finding of a better consistency in performance (measured as reduced intraset power fluctuations) at the end of a dynamic repeated effort.

KEY WORDS high-intensity efforts, detraining, power maintenance, neuromuscular performance

INTRODUCTION

In most team sports, players are required to repeatedly perform brief, short, high-intensity actions (HIA), such as sprints, accelerations, decelerations, changes of direction, and jumps (3,16). As such, this ability to repeat HIA throughout a match has been identified as an important fitness prerequisite in such sports (7). In addition to lower-body explosive actions, in most team sports (e.g., basketball, handball, rugby), players have to perform high-intensity total body actions. Because such ability has been linked to performance and discrimination between player's competitiveness in basketball (2,27), it appears as an area of interest for team sports strength and conditioning coaches.

However, substantial decrements in the frequency of HIA have been reported at the latter stages of a match (e.g., second half) (3,23), and several training strategies (e.g., high-intensity interval training, repeated sprints, speed endurance, and plyometrics) have been recommended to enhance HIA maintenance during lower-body actions in team-sports athletes (4,6). With training time at a premium, the selection and search of training strategies that are able to concurrently target a wide variety of adaptations (e.g., metabolic, mechanical, neuromuscular) is essential. As such, several blocks of sets of lower-body maximal power training with incomplete recovery periods in between sets (i.e., repeated power ability [RPA]; a term originally coined by Tous-Fajardo and Moras (29)) have been recently proposed as a highly effective training modality to improve fatigue resistance during HIA (i.e., repeated-sprint exercise) (15,28). Nevertheless, no data are currently available on the effect of any upper-body training strategy over the maintenance of upper-body HIA (e.g., throwing, passing, blocking).

Based on the scarce information about upper-body repeated efforts, a reliable and useful upper-body RPA test has been proposed to monitor short- and long-term changes in power output maintenance during the bench press exercise (14). Despite that a training program including 3 arm-curl sets to failure with 1-minute recovery intervals has shown a higher maintenance of average repetition

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velocity than training with longer recovery intervals (4 minutes) (11), there are no studies, to our knowledge, which have analyzed the impact of repeating several blocks of sets of few reps at the optimal load (i.e., the load that maximizes power output) with very short recovery periods that are closer to the real-game demands (e.g., 30 seconds) (2,3) in both the load-power spectrum and the power maintenance. Thus, it is possible that this type of training might improve the ability to maintain performance during repeat high-intensity upper-body actions in the later stages of the game when fatigue might be a limiting factor (4).

A recent meta-analysis has shown an inappreciable decrement in muscle power after a training cessation of 28 days (5). However, to date, neither the effect of RPA training and training cessation nor the effect of any training over RPA has been evaluated. Thereby, there is a need to understand the time course and perdurability of RPA adaptations to draw an adequate training periodization. Therefore, the aims of this study were (a) to analyze the impact of a RPA training on different strength, power, and repeated power measurements and (b) to examine the effect of a training cessation period on the ability to repeat power and fatigue measurements.

METHODS

Experimental Approach to the Problem

Using a controlled study design, participants were allocated randomly (ABBA method) into a control group (CON; $n = 10$; 1 repetition maximum [1RM] = 59.4 ± 8.8 kg) or a repeated power group (RPG; $n = 10$; 1RM = 59.7 ± 10.7 kg) based on their RPA performance (i.e., average power over 5 sets) (15). Both groups continued performing their normal and identical strength training. Such training was compounded by lateral pull-down, shoulder military press, and seated cable row (4–8 repetitions, 80–100% of the maximal power load with 3' of recovery between sets). Exclusively, the only difference between both groups was the execution of the bench press exercise. Although CON group did not perform the bench press exercise during the study period, RPG performed the RPA training in the bench press exercise. Participants within each group were matched according to their initial performance in the average power over 5 sets (AP_M) on RPA test. Tests were performed in a gym, 2 weeks and 1 week (test-retest) before the commencement of the training and 1 week after finishing the intervention. Subsequently, a training cessation period was included, with players tested 4 weeks after the last assessment (postcessation). Neither CON nor RPG participated in strength training during this particular period. Tests included a 1RM test, an incremental load test, and a RPA test on the bench press exercise. Players were familiarized with the exercise procedures before starting testing sessions.

Subjects

Twenty young (U-15 to U-17), highly trained, male basketball players were selected ($\pm SD$ age: 15.9 ± 0.9 years, height: 189.1 ± 9.8 cm, body mass: 81.3 ± 8.9 kg). Data collection took place in the second month of the competitive season after a 2-month preseason period. All players had at least 5 years of experience and participated on average in ~ 10 hours of combined basketball (5–6 sessions) and strength/power (2 sessions) and 2 competitive matches per week. At the time of the study, all players were competing at a national level (i.e., Spanish Basketball National League), with 6 of them also competing at an international level (i.e., U-15 to U-17 European and World Basketball Championships). Every subject had a minimum of 1-year training experience with the bench press exercise. All procedures performed were in accordance with the ethical standards of the San Jorge University Ethics Committee and with the 1964 Helsinki declaration. Informed written consent was obtained from all individual participants and his or her parents included in the study.

Procedures

Training Intervention. Subjects performed 2 training sessions per week, before the technical-tactical sessions (18:00–20:00 hours) during 7 weeks. At least, 48 recovery hours were provided between each RPA session. Repeated power ability training consisted of 1–3 blocks of 5 sets $\times$ 5 repetitions using the load that maximized power output ($Load_{opt}$) on the bench press exercise. Recovery period was 30 seconds between sets and 3 minutes between blocks. Both recoveries were passive. Training was periodized in a progressive manner: 1 (weeks 1–2), 2 (weeks 3–4), or 3 (weeks 5–6) blocks of 5 sets $\times$ 5 repetitions were performed with 30 seconds and 3 minutes of passive recovery between each set and block, respectively. A decrement taper was performed during the seventh week; 2 and 1 sets were completed during the first and second session in the taper period, respectively. The eccentric phase was performed at a moderate velocity (i.e., self-selected and never exceeding 3 seconds), whereas the concentric phase as fast as possible. The main researcher controlled every training session. Each training session was preceded by a 15-minute standardized warm-up, which included jogging, arm and shoulder mobilization, 1 set of 8 repetitions at 40% of 1RM, 1 set of 4 repetitions at 60% of 1RM, and 1 set of 3 repetitions at $Load_{opt}$ (i.e., specific load), all performed in an explosive manner in the bench press exercise. After warming up, subjects rested 5 minutes before starting the RPA training. The selection of 5 repetitions per set was established because these participants were able to maintain their maximum power output (MPO; above 90% of MPO) during approximately 5 repetitions in pilot studies.

One Repetition Maximum Assessment. At least 72 hours before beginning the incremental load test, the 1RM test in the bench press exercise was conducted as previously described (17).

TABLE 1. Within-group differences in the incremental load test in the bench press exercise in the repeated power group and control group between pretraining and posttraining.*

Variables	RPG (<i>n</i> = 10)						CON (<i>n</i> = 10)					
	Pre	Post	% Difference (90% CL)	Standardized difference (90% CL)	Qualitative Chances assessment		Pretest	Posttest	% Difference (90% CL)	Standardized difference (90% CL)	Qualitative Chances assessment	
1RM (kg)	59.7 ± 10.2	62.8 ± 12.0	4.8 (1.4 to 8.4)	0.26 (0.08 to 0.45)	72/28/0%	Possibly	59.4 ± 8.8	57.6 ± 8.9	-3.2 (-6.4 to 0.2)	-0.20 (-0.42 to 0.01)	0/48/51%	Possibly poorer
MPO (W)	421.5 ± 87.8	472.2 ± 93.8	12.1 (6.3 to 18.1)	0.54 (0.29 to 0.78)	98/2/0%	Very likely	409.3 ± 72.8	441.7 ± 72.0	8.1 (5.2 to 11.0)	0.41 (0.27 to 0.55)	99/1/0%	Very likely
Load _{opt} (kg)	27.7 ± 5.2	28.3 ± 5.4	2.2 (-0.5 to 5.0)	0.11 (-0.02 to 0.24)	12/88/0%	Likely trivial	27.5 ± 3.7	25.6 ± 4.4	-7.5 (-14.1 to -0.3)	-0.54 (-1.05 to -0.02)	1/12/87%	Likely poorer
AP ₂₀ (W)	323.2 ± 61.5	372.2 ± 77.4	14.7 (7.6 to 22.2)	0.73 (0.39 to 1.07)	99/1/0%	Almost certain	333.3 ± 63.0	361.8 ± 51.4	9.3 (2.7 to 16.3)	0.44 (0.13 to 0.76)	91/9/0%	Likely
AP ₄₀ (W)	417.2 ± 96.4	451.6 ± 79.8	9.2 (3.7 to 15.0)	0.37 (0.15 to 0.98)	91/9/0%	Likely	400.1 ± 76.0	439.6 ± 75.3	10.2 (4.5 to 16.1)	0.47 (0.22 to 0.73)	96/4/0%	Very likely
AP ₆₀ (W)	413.8 ± 94.5	461.4 ± 96.3	11.7 (3.9 to 20.2)	0.48 (0.16 to 0.79)	93/7/0%	Likely	399.2 ± 65.4	416.1 ± 78.6	3.7 (-3.1 to 10.9)	0.20 (-0.18 to 0.58)	51/45/4%	Possibly
AP ₈₀ (W)	322.4 ± 61.9	393.4 ± 95.8	20.8 (9.9-32.8)	0.89 (0.44 to 1.33)	99/1/0%	Almost certain	332.2 ± 49.4	326.5 ± 44.8	-1.6 (-8.7 to 6.1)	-0.10 (-0.57 to 0.37)	14/51/35%	Unclear

*RPG = repeated power group; CON = control group; %Difference = percentage difference; Standardized difference = effect size; CL = confidence limit; Chances = percentage chance of having better/similar/poorer values; 1RM = 1 repetition maximum; MPO = maximum power output; Load_{opt} = load that maximizes maximum power; AP₂₀ = average power at 20% of 1RM; AP₄₀ = average power at 40% of 1RM; AP₆₀ = average power at 60% of 1RM; AP₈₀ = average power at 80% of 1RM.

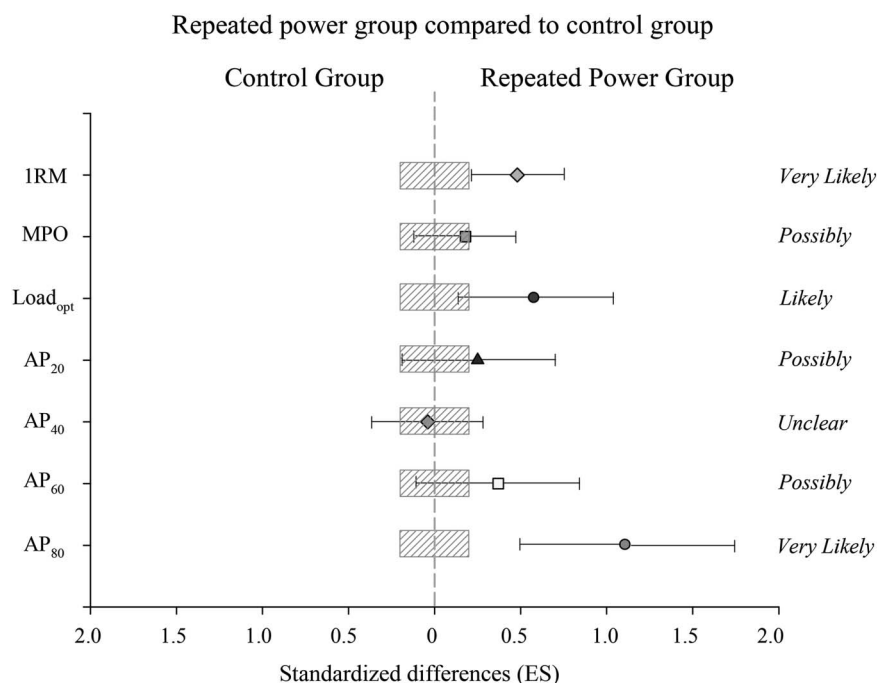


Figure 1. Efficiency of the repeated power group (RPG) in comparison with control group (CON) to improve 1 repetition maximum (1RM), maximum power output (MPO), the load that maximized power output (Load_{opt}), and average power at 20, 40, 60, and 80% of 1RM (bars indicate uncertainty in the true mean changes with 90% confidence intervals). Trivial areas were the smallest worthwhile change (see Methods). AP₂₀ = average power at 20% of 1RM; AP₄₀ = average power at 40% of 1RM; AP₆₀ = average power at 60% of 1RM; AP₈₀ = average power at 80% of 1RM.

Incremental Load Test in Bench Press. An incremental load test was used to determine the Load_{opt}. Four loads (20, 40, 60, and 80% of 1RM) were used to determine the force-velocity relationship as described elsewhere (14). Estimated 1RM, MPO, and Load_{opt} were calculated by specific software (Musclelab V.8, Langesund, Norway). Intraclass correlation coefficient and coefficient of variation (CV, %) values in all loads and variables ranged from 0.94 to 0.99, and from 2.2 to 3.7%, respectively.

Repeated Power Ability Test. A test that consisted of 5 sets of 5 repetitions with the Load_{opt}, with 30 seconds of passive recovery between sets was employed as described elsewhere (14). The variables used for the analysis were average power (AP) of each set, AP_M, RPA index 1 (AP in the last set [AP_L] at posttraining/AP in the best set [AP_B] at pretraining), RPA index 2 (AP_L at postcessation/AP_B at pretraining), RPA index 3 (AP_L at postcessation/AP_B at posttraining), and percentage of power decrement (%Dec) across the 5 sets. Percentage of power decrement was calculated using the following formula: %Dec = (25/[total power across 5 sets/5])/(25/[total power of 4 and 5 set]/2) × 100. This formula was chosen because it was the most reliable (CV: 1.7% [CL 90%: 1.4–2.1]) of those proposed by Glaister et al. (13), in our own pilot studies. Furthermore, a new fatigue-related

variable, intraset power fluctuation (FLUC), was introduced and calculated as follows: FLUC = (SD of AP in each set/mean of each set) × 100. Intraclass correlation coefficient values for each set (i.e., AP) were between 0.95 and 0.98, and CV values were between 3.4 and 5.6% (14).

Statistical Analyses

Data are presented as mean ± SD. All data were firstly log-transformed to reduce bias arising from nonuniformity error. A Shapiro-Wilk test was done to assess if the data were normally distributed. All data were normally distributed. To analyze the between-set differences in the RPA test, a 2-way (treatment × time) analysis of variance for repeated measurements was performed with Bonferroni adjustments to execute the pairwise comparisons when needed. Subsequently, effect size (ES) calculation and threshold values (confidence limits [CLs] 90%) were established as described elsewhere (magnitude-based inferences approach) (18). For within-group/between-group comparisons, the chances that the differences in performance were better/greater (i.e., greater than the smallest worthwhile change [0.2 multiplied by the between-subject SD, based on Cohen's *d* principle]) similar or worse/smaller were calculated. Quantitative chances of beneficial/better or detrimental/poorer effect were assessed qualitatively as follows: <1%, almost certainly not;

TABLE 2. Within-group differences in the repeated power ability test in the bench press exercise in the repeated power group and control group between pretraining and posttraining.*

Variables	RPG (<i>n</i> = 10)					CON (<i>n</i> = 10)				
	Pre	Post	% Difference (90% CL)	Standardized difference (90% CL)	Qualitative Chances assessment	Pretest	Posttest	% Difference (90% CL)	Standardized difference (90% CL)	Qualitative Chances assessment
AP _B (W)	393.1 ± 76.7	466.5 ± 89.9	18.7 (13.2 to 24.4)	0.84 (0.60 to 1.07)	100/0/0% Almost certain	398.3 ± 69.5	413.8 ± 71.7	3.8 (0.5 to 7.3)	0.20 (0.02 to 0.37)	48/52/0% Possibly
AP _L (W)	354.2 ± 74.2	404.7 ± 74.3	14.6 (8.3 to 21.2)	0.62 (0.36 to 0.88)	99/1/0% Almost certain	354.3 ± 80.2	338.8 ± 71.8	-4.3 (-8.0 to -0.3)	-0.18 (-0.36 to -0.01)	0/56/44% Possibly trivial
AP _M (W)	371.6 ± 77.3	426.4 ± 79.8	15.1 (10.1 to 20.3)	0.65 (0.45 to 0.85)	100/0/0% Almost certain	374.0 ± 72.9	371.0 ± 67.9	-0.7 (-3.9 to 2.6)	-0.03 (-0.19 to 0.12)	1/95/4% Very likely trivial
%Dec	95.7 ± 1.6	96.0 ± 2.4	0.3 (-0.8 to 1.3)	0.15 (-0.41 to 0.72)	44/42/14% Unclear	95.0 ± 4.4	93.1 ± 4.6	-2.1 (13.4 to 377.6)	-0.42 (-0.79 to -0.05)	1/15/85% Likely poorer
FLUCB	5.8 ± 3.2	5.7 ± 1.8	2.4 (-31.9 to 53.8)	0.04 (-0.66 to 0.74)	34/39/27% Unclear	5.3 ± 1.5	5.0 ± 2.1	7.6 (-18.4 to 27.9)	0.26 (-0.56 to 1.09)	55/28/16% Unclear
FLUCL	7.6 ± 2.9	5.3 ± 1.3	37.3 (-5.3 to 99.0)	1.09 (-0.19 to 2.38)	88/7/5% Likely	9.3 ± 3.1	9.2 ± 3.2	1.8 (-12.2 to 14.1)	0.05 (-0.33 to 0.44)	25/62/13% Unclear
FLUCM	7.0 ± 1.5	6.1 ± 1.1	14.0 (-4.0 to 35.2)	0.62 (-0.19 to 1.43)	82/13/5% Likely	7.8 ± 1.8	7.1 ± 1.3	8.2 (-1.2 to 16.7)	0.32 (-0.04 to 0.68)	72/26/1% Possibly

*RPG = repeated power group; CON = control group; %Difference = percentage difference; Standardized difference = effect size; CL = confidence limit; Chances = percentage chance of having better/similar/poorer values; AP_B = average power in the best set; AP_L = average power in the last set; AP_M = average power over the 5 sets of the repeated power ability test; %Dec = percentage of decrement; FLUC_B = power fluctuations in the best set; FLUC_L = power fluctuations in the last set; FLUC_M = power fluctuations in the mean of 5 sets in the repeated power ability test.

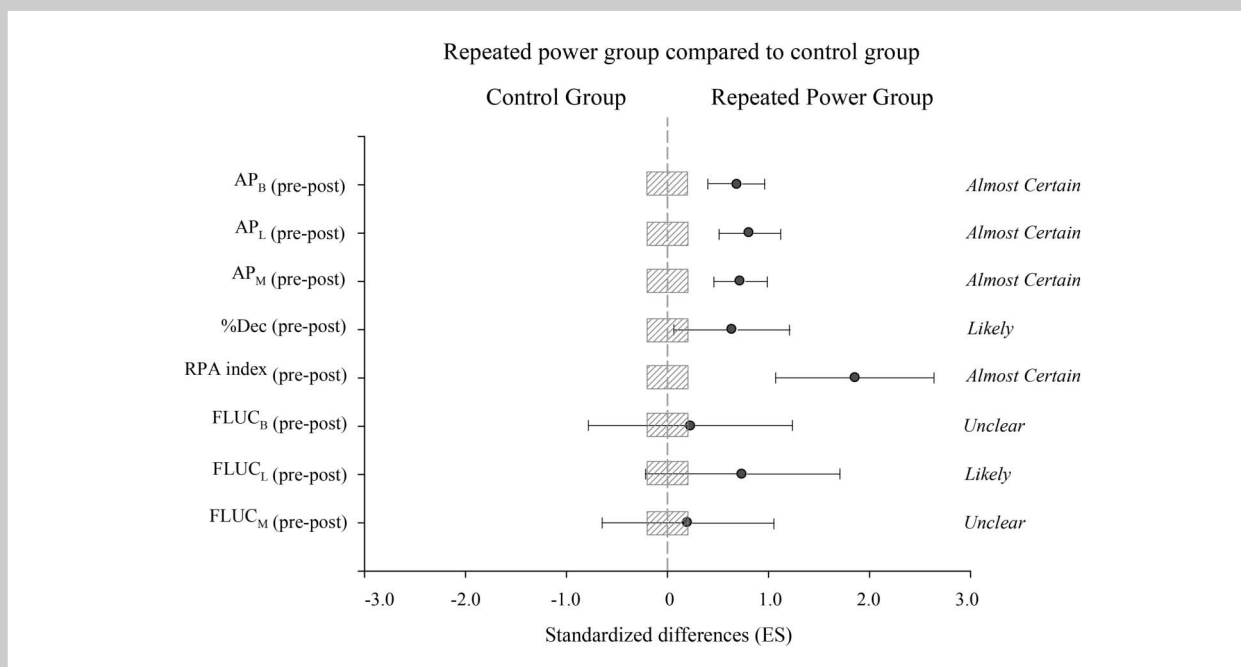


Figure 2. Efficiency of the repeated power group (RPG) in comparison with control group (CON) to improve average power in the best set (AP_B pre-post), average power in the last set (AP_L pre-post), average power over the 5 sets between pretraining to posttraining (AP_M pre-post), the percentage of decrement (% Dec pre-post), the RPA index (RPA index pre-post), the power fluctuations in the best ($FLUC_B$ pre-post), last set ($FLUC_L$ pre-post) and the mean of 5 sets ($FLUC_M$ pre-post) between the pretraining and posttraining in the repeated power ability test (bars indicate uncertainty in the true mean changes with 90% confidence intervals). Trivial areas were the smallest worthwhile change (see Methods).

>1–5%, very unlikely; >5–25%, unlikely; >25–75%, possible; >75–95%, likely; >95–99%, very likely; and >99%, almost certain (18). If the chance of having beneficial/better or detrimental/poorer performances was both >5%, the true difference was assessed as unclear. Otherwise, we interpreted that change as the observed chance (18). The Pearson product moment correlation coefficient was used to determine the relationship between different variables and improvements. The following criteria were adopted to

interpret the magnitude of the correlation (r) between variables: ≤ 0.1 , trivial; >0.1–0.3, small; >0.3–0.5, moderate; >0.5–0.7, large; >0.7–0.9, very large; and >0.9–1.0, almost perfect (18).

RESULTS

One Repetition Maximum Test

Baseline bench press 1RM was 59.6 ± 9.3 kg for the whole group (CON and RPG).

TABLE 3. Repeated power ability indexes in the bench press exercise in the repeated power group and control group at different times.*

Variables	RPG ($n = 10$)	CON ($n = 10$)	Difference (90% CL)	Standardized difference (90% CL)	Chances	Qualitative assessment
RPAindex 1	1.03 ± 0.10	0.85 ± 0.08	21.9% (13.2–31.4)	1.98 (1.24–2.73)	100/0/ 0%	Almost certain
RPAindex 2	0.99 ± 0.08	0.82 ± 0.05	21.4% (14.9–28.4)	2.60 (1.85–3.34)	100/0/ 0%	Almost certain
RPAindex 3	0.83 ± 0.05	0.79 ± 0.07	6.2% (0.4–12.4)	0.80 (0.06–1.55)	91/7/ 2%	Likely

*RPG = repeated power group; CON = control group; CL = confidence limit; RPAindex 1 = AP_L at posttraining/ AP_B at pretraining; RPAindex 2 = AP_L at postcessation/ AP_B at pretraining; RPAindex 3 = AP_L at postcessation/ AP_B at posttraining; AP_B = average power in the best set during the repeated power ability test; AP_L = average power in the last set during the repeated power ability test.

TABLE 4. Within-group differences in the repeated power ability test in the bench press exercise in the repeated power group and control group between posttraining and postcessation.*

Variables	RPG (<i>n</i> = 10)					CON (<i>n</i> = 10)				
	Post	Post-4 week	% Difference (90% CL)	Standardized difference (90% CL)	Qualitative Chances assessment	Post	Post-4 week	% Difference (90% CL)	Standardized difference (90% CL)	Qualitative Chances assessment
AP _B (W)	466.5 ± 89.9	458.5 ± 96.0	-1.9 (-5.1 to 1.3)	-0.09 (-0.25 to 0.06)	0/88/12% Likely trivial	413.8 ± 71.7	399.0 ± 62.5	-3.3 (-5.8 to -0.7)	-0.17 (-0.30 to -0.04)	0/66/34% Possibly trivial
AP _L (W)	404.7 ± 74.3	390.9 ± 89.9	-4.1 (-7.6 to -0.4)	-0.20 (-0.38 to -0.02)	0/50/50% Possibly trivial	338.8 ± 71.8	321.6 ± 61.0	-4.7 (-9.7 to 0.5)	-0.21 (-0.43 to 0.02)	0/48/52% Possibly poorer
AP _M (W)	426.4 ± 79.8	421.7 ± 92.6	-1.7 (-4.3 to 1.1)	-0.08 (-0.22 to 0.05)	0/93/7% Likely trivial	371.0 ± 67.9	356.7 ± 63.1	-3.7 (-6.8 to -0.6)	-0.18 (-0.34 to -0.03)	0/57/43% Possibly trivial
%Dec	96.0 ± 2.4	95.2 ± 1.3	-0.8 (-2.4 to 0.8)	-0.29 (-0.87 to 0.29)	8/31/61% Unclear	93.1 ± 4.6	93.3 ± 2.0	0.3 (-2.2 to 2.9)	0.06 (-0.42 to 0.53)	29/53/17% Unclear
FLUC _B	5.7 ± 1.8	5.5 ± 2.1	10.8 (-20.3 to 54.1)	0.21 (-0.46 to 0.88)	51/35/14% Unclear	5.0 ± 2.1	6.3 ± 2.3	-23.9 (-64.6 to 6.7)	-0.56 (-1.30 to 0.18)	5/15/80% Likely poorer
FLUC _L	5.3 ± 1.3	8.0 ± 4.0	-39.7 (-128.8 to 14.6)	-1.12 (-2.77 to 0.53)	9/8/84% Likely poorer	9.2 ± 3.2	9.7 ± 3.2	-6.1 (-42.8 to 21.2)	-0.15 (-0.91 to 0.61)	21/34/45% Unclear
FLUC _M	6.1 ± 1.1	6.3 ± 1.7	-1.7 (-20.7 to 14.3)	-0.07 (-0.82 to 0.67)	25/37/38% Unclear	7.1 ± 1.3	7.3 ± 2.1	-0.6 (-14.5 to 11.6)	-0.03 (-0.64 to 0.58)	25/44/31% Unclear

*RPG = repeated power group; CON = control group; %Difference = percentage difference; CL = confidence limit; Standardized difference = effect size; Chances = percentage chance of having better/similar/poorer values; AP_B = average power in the best set; AP_L = average power in the last set; AP_M = average power over the 5 sets of the repeated power ability test; %Dec = percentage of decrement; FLUC_B = power fluctuations in the best set; FLUC_L = power fluctuations in the last set; FLUC_M = power fluctuations in the mean of 5 sets in the repeated power ability test.

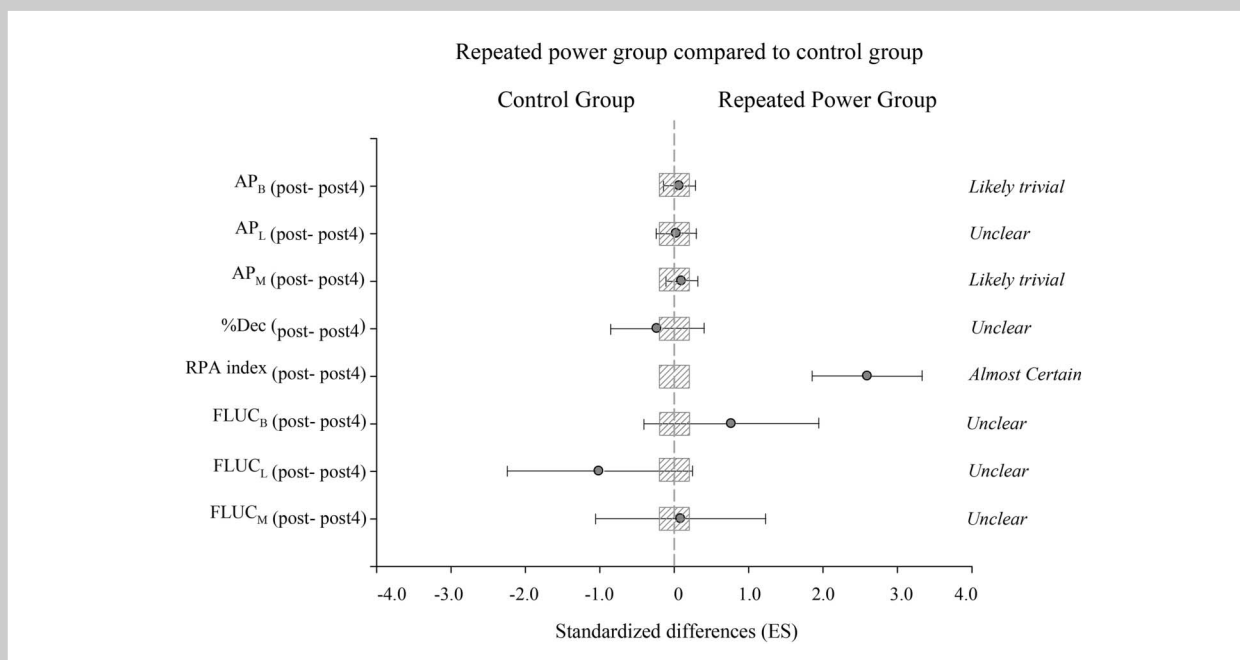


Figure 3. Efficiency of the repeated power group (RPG) in comparison with control group (CON) to improve average power in the best set (AP_B posttraining to postcessation), average power in the last set (AP_L posttraining to postcessation), average power over the 5 sets (AP_M posttraining to postcessation), and the percentage of decrement (%Dec posttraining to postcessation), the RPA index post-4 (RPA index posttraining to postcessation), the power fluctuations in the best (FLUC_B posttraining to postcessation), last set (FLUC_L posttraining to postcessation) and the mean of 5 sets (FLUC_M posttraining to postcessation) between the posttraining and the final of the training cessation period in the repeated power ability test (bars indicate uncertainty in the true mean changes with 90% confidence intervals). Trivial areas were the smallest worthwhile change (see Methods).

Incremental Load Test

MPO, AP at 20% 1RM (AP₂₀) and 40% 1RM (AP₄₀) were substantially enhanced in both treatments in the posttraining compared with the pretraining. Furthermore, substantial greater values were found in AP at 60% 1RM (AP₆₀) and 80% 1RM (AP₈₀) in RPG after the training period (posttraining in comparison with pretraining). Within-group results are shown in Table 1. Between-treatment comparisons are illustrated in Figure 1. Substantially better 1RM (% = 8.2%, [CL 90%: 3.5–13.2], with chances for greater/similar/lower performance of 95/5/0%), Load_{opt} (% = 10.5%, [CL 90%: 2.3–19.4], 92/7/0%), and AP₈₀ (% = 22.8%, [9.5–37.6], 99/1/0%) was found in RPG in comparison with CON after the training period. Substantially greater AP₄₀ and AP₆₀ were found (*almost certainly*) at pretraining in comparison with AP₂₀ and AP₈₀ in both treatments. Interestingly, AP₂₀ was substantially better than AP₈₀ at posttraining (ES = 0.67, [CL 90%: 0.14–1.20], 93/6/1%) in CON, whereas a *possibly* better performance in AP₈₀ than AP₂₀ (ES = 0.21, [–0.20 to 0.69], 52/43/5%) was found in RPG at posttraining.

Repeated Power Ability Test

Inter-set analysis showed significant differences between AP_B with AP₄ ($p = 0.002$) and AP_L ($p = 0.002$) and AP₃ with AP₄ ($p = 0.039$) and AP_L ($p = 0.021$) at pretraining, whereas significant differences were only observed between AP_B

and AP₂ ($p = 0.012$), AP₃ ($p = 0.001$), AP₄ ($p = 0.001$), and AP_L ($p = 0.001$) at posttraining. Substantially better results were found in AP_B, AP_L, and AP_M in RPG between pretraining and posttraining. Interestingly, a *likely poorer* %Dec was reported in CON at posttraining, whereas it remained unchanged (*unclear*) in RPG (Table 2). FLUC during the last set (FLUC_L) and in the mean of 5 sets (FLUC_M) were substantially improved in RPG, whereas no substantial changes were observed at any other set in any of the groups. Furthermore, no substantial changes were reported in CON between pretraining and posttraining.

Between-group differences showed substantial differences between pretraining and posttraining in AP_B, AP_L, AP_M, and %Dec in RPG in comparison with CON (Figure 2). The RPG displayed a substantial better RPA index at posttraining (%: 21.9 [CL 90%: 13.2–31.4], 100/0/0%) (Table 3). Furthermore, a substantial reduction in FLUC_L was found in RPG in comparison with CON (%: 25.8, [CL 90%: –9.1 to 49.5], 84/11/5%) between pretraining to posttraining.

Training Cessation

No substantial differences were established within any of the 2 groups (Table 4), with the exception of FLUC_L and FLUC_B in RPG and CON, respectively. No between-group differences were found in AP_B, AP_L, AP_M, and %Dec between posttraining and postcessation (Figure 3). The

RPG displayed a substantial better RPA index than CON at postcessation (%: 21.4 [CL 90%: 14.9–28.4], 100/0/0%).

Relationships Between Strength Performances Indices

When data from both groups were pooled, the improvement of 1RM pretraining to posttraining was *largely* correlated to the enhancement of AP_B pretraining to posttraining ($r = 0.61$; CL 90%: 0.30–0.80), AP_L pretraining to posttraining ($r = 0.61$; CL 90%: 0.30–0.80), and AP_M pretraining to posttraining ($r = 0.65$; CL 90%: 0.37–0.83). A *very large* relationship was found between the improvement of AP_B pretraining to posttraining and the improvement of AP_L pretraining to posttraining ($r = 0.83$; CL 90%: 0.65–0.92). Interestingly, the improvement of AP_B pretraining to posttraining and the improvement of AP_M pretraining to posttraining were *almost perfectly* correlated ($r = 0.91$; CL 90%: 0.82–0.96). Furthermore, RPA index post was *very largely* correlated with the pretraining to posttraining improvements in AP_B ($r = 0.84$; CL 90%: 0.68–0.99), AP_L ($r = 0.84$; CL 90%: 0.68–0.99), and AP_M ($r = 0.85$; CL 90%: 0.69–0.99) in RPG. No substantial correlations were found between RPA index post and any change in FLUC, with the exception of FLUC_L ($r = 0.53$; CL 90%: 0.13–0.93) in RPG. Substantial correlations were also established between decrement in FLUC_L and the improvement in AP_L ($r = 0.60$; CL 90%: 0.25–0.96) between pretraining to posttraining in RPG. In addition, changes in FLUC_B were very largely correlated ($r = 0.72$; CL 90%: 0.50–0.94) with those in FLUC_M.

DISCUSSION

The aims of the present study were to analyze the addition of a RPA training to habitual strength training sessions on the bench press RPA test, 1RM, and load-power spectrum tests and to describe the effects of training cessation. The main findings were as follows: (a) 7 weeks of repeated maximal power training induced *moderate to large* magnitude improvements in the RPA test, (b) training group achieved greater results in the worst set at posttraining than the best effort during pretraining (RPA index), (c) power fluctuations during the last set were substantially improved (i.e., less interrepetition variability) in RPG in comparison with controls, (d) substantial improvements were found in the incremental load test (i.e., 1RM, Load_{opt}, 80% 1RM) in RPG in comparison with CON, and (e) RPA performance (AP_B, AP_L, and AP_M) was maintained after 4 weeks of training cessation in both RPG and CON.

This is the first study analyzing the effect of a RPA training on upper-body performance showing a substantial improvement over the full RPA spectrum (i.e., AP_B, AP_L, AP_M, RPA index, and FLUC_L). Despite all participants in RPG enhanced their RPA performance (i.e., AP_M), an intersubject variability was observed (5.1–30.5% improvements). This could be because of the initial strength level that has been reported to be inversely related to training-induced improvements in strength and power (9). However, the relationship

between 1RM at pretraining and the enhancement in AP_M pretraining to posttraining was *small*, meaning that moderately trained subjects may improve their upper-body RPA irrespective of their strength training background and/or status (weaker/stronger).

Despite that participants performed a reliability analysis, a learning effect (i.e., improvement in test-specific coordination) is very likely to occur with training repetition (20). Furthermore, a progressive repeated-effort enhancement has been observed when a repeated-effort test is administered over time (12). Hence, it is possible that the effectiveness of RPA training on RPA performance might have been overestimated. Nevertheless, the *large* relationship between the 1RM increase and the AP_M improvement at posttraining in addition to a *possibly* better 1RM performance in RPG indicates that additional factors need to be considered. For example, it has been suggested that several neuromuscular adaptations, such as a better elastic energy use (1) and contractile elements interaction (8), an increase in fascicle length, a higher motor unit (MU) recruitment (24) and firing frequency, an optimal MU synchronization or a better intermuscular coordination (8), have been elicited in addition to, or rather than, a potential learning effect. Further studies are needed to better understand what neuromuscular and metabolic adaptations are triggered through RPA training.

Previously, it has been reported that the main RPA determinant is AP_B (14), which agrees with the present results showing this strong relationship at both after training and after 4 weeks of training cessation. In addition, the improvements in AP_B and AP_M were *almost perfectly* correlated, perhaps suggesting that conventional power training at the optimal load would enhance RPA performance. However, despite both groups improved MPO, only the intervention group substantially improved AP_B. Given the *almost certainly* differences between groups in AP_B, it seems that the best repetition (MPO) and the best set (AP_B) might depend on different factors. In this regard, because CON performed several maximal power upper-body exercises (i.e., lateral pull-down, shoulder press, and cable row) avoiding the bench press, it is hypothesized that MPO was enhanced through greater explosive force production rather than better intermuscle coordination as might occur in RPA performance.

One of the most interesting and novel findings was the substantial greater improvement of RPA index in RPG compared with CON. Furthermore, a very large correlation was found between RPA index post and improvements in AP_B, AP_L, and AP_M. From a practical perspective, this would mean that a player would be able to perform similarly to the best action before training (AP_B pretraining) after several actions (AP_L posttraining) throughout a game.

To the best of our knowledge, this is the first study showing a decrease in the variability of intraset power output, specifically during the last set (FLUC_L) of the RPA test. Despite it is well-known that neuromuscular fatigue

increases the force output fluctuations, most of the literature is based on intrarep analysis during sustained isometric actions until a target level can no longer be maintained (21,22,30). Interestingly, a study by Tracy et al. (30) showed in old adults that a heavy-load strength training reduced the intrarep force fluctuations during anisometric actions but not during isometric actions. However, our study monitored intraset power fluctuations, and comparisons cannot be made because it is uncertain to what extent both fluctuations' behavior rely on the same mechanisms. Nevertheless, the intraset analysis appears of value because large correlations were found between decrements in $FLUC_L$ and both RPA index post and improvements in AP_L , perhaps meaning that a better consistency within a set when fatigue effects appear can help in maintaining power output. Given that significant decreases in power output were not found between all sets at pretraining, a more demanding RPA test protocol (e.g., 15-second rest intervals instead of 30 seconds) might be needed to assess the impact of fluctuations behavior. Furthermore, such protocol should be proposed in well-trained subjects to obtain greater fatigue indexes and thus expect even greater fatigue resistance adaptations. In any case, it should be noted that at posttraining no significant differences in power output were found from the second to the fifth set at training group, whereas controls increased their between-set differences. As such, this finding may legitimate this training strategy to improve power consistency after accumulated sets. Collectively, and in addition to the unchanged (*unclear effect* or *likely* better than CON) %Dec, it seems that RPA training might induce substantial fatigue resistance adaptations that clearly deserve further research.

However, despite no substantial changes were found in 1RM in any group, CON showed a *small* impairment ($ES = -0.20$) in contrast with a *possibly* ($ES = 0.26$) better performance in RPG. Interestingly, MPO, AP_{20} , and AP_{40} were substantially enhanced in CON, perhaps because of the previously mentioned inclusion of maximal velocity upper-body exercises with similar loads to those in the RPG (38–45% 1RM). These exercises might influence the present results and underestimate between-group differences, and consequently, it should be considered as a limitation. Also, it might be possible that intramuscular adaptations (i.e., firing frequency, MU recruitment) helped to maintain or even improve performance with loads that maximized MPO despite different exercises were performed. Thus, it seems that a range of loads surrounding MPO might be more appropriate than specific exercise to develop power adaptations with light loads (20 and 40% 1RM) in young basketball players. Anyway, AP_{20} and AP_{80} were the loads more largely improved in RPG, which is in agreement with the results found in a study including a group with similar 1RM values, where a maximal power training produced the greatest improvements with the lowest (35% 1RM) and the highest (95% 1RM) loads (19). Thus, it is supported that young basketball players or inexperienced subjects (19) might not

need high training loads to enhance maximal strength and AP in the full range of loads. In reference to MPO, the present results are also within the range ($ES = 0.37$ – 0.86) reported in similar participants (10,19) to our players ($ES = 0.54$).

One of the most important findings was the maintenance of the RPA performance (i.e., AP_M) 4 weeks after completing the training program. This is in agreement with other studies (25,26), which have found no significant changes in power/strength or even improvements after different training cessation periods in young basketball players. A recent meta-analysis (5) also reported no substantial changes in maximal power after a similar period (i.e., 4 weeks). Conversely, AP_L was *possibly* impaired in CON after postcessation and a slightly greater decrement in RPA performance in comparison with RPG at postcessation was manifested. Therefore, it seems that only the specific exercise and training might be able to maintain adaptations after a strength training cessation period.

Future studies should be conducted to compare the effects of a RPA training with other training methods. Also, it would be interesting to include a modified upper-body functional test such as a throwing or a medicine ball explosive test (25,26), where the same repeated efforts are performed.

PRACTICAL APPLICATIONS

The addition of a twice-per-week RPA training represents an effective means to enhance a wide variety of adaptations, including maximal strength (1RM), MPO, maximal power with low, medium, and high loads, and the ability to repeat power in the upper body. Interestingly, training adaptations appeared more robust in those variables related to fatigue resistance, and for the first time, a better consistency in performance (measured as reduced intraset power fluctuations) was proposed as a potential determinant of power maintenance at the end of a dynamic repeated effort. Furthermore, repeated lifting of moderate loads at high speeds might prompt neural adaptations and influence directly in high loads without using those loads. As such, a new perspective is opened through the inclusion of this training paradigm to also improve the dynamic maximal strength of young and/or moderately strength trained basketball players. Finally, those coaches who want to focus on technical or tactical concepts previously to a tournament or championship (i.e., within 4 weeks) might prescribe (maintain or cease) their power training without compromising its RPA performance.

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